

LAB: 01

Hexadecimal Analyzing Digital Evidence

Objective:

- To understand that files have unique file headers based in type of file
- Installing Autopsy
- To Analyze files on USB drive

Activities:

Task 1

Hexadecimal

Note: Please include a screenshot of each step. Crop your screenshot to show relevant information only. Uncropped screenshots will result in a 10% deduction from your marks.

Learning Activities:

At the end of these activities, you should understand:

- How to determine key details related to basic file headers.
- How to search and generate report using Autopsy.

1) Download and Install Winhex software on your computer.

(<http://www.winhex.com/winhex/hex-editor.html>)

2) Download the Lab1_files.zip (week2) from DC Connect.

3) Open each file in Winhex.

4) Using a chart of file signatures determine what type the file are listed in the following URL and fill the following table:

http://en.wikipedia.org/wiki/List_of_file_signatures

File_Name	Hex Signature	File extension	offset	Description
File1	50 4B 03 04	.docx	0	PK
File2	25 50 44 46 2D	.pdf	0	%PDF-
File3	42 4D FE B3	.bmp	0	BMp ³
File4	47 49 46 38 39 61	.gif	0	GIF89a
File5	D4 C3 B2 A1	.pcap	0	ÔÃ²;
File6	25 50 44 46 2D	.txt	0	This file was changed from pdf to txt by changing extension, it was

				not converted to text format: %PDF-
File7	47 49 46 38 39 61	.pdf	0	File is originally a gif file, but extension was changed to pdf: GIF89a
File8	-	.txt	-	Text files don't have Hex signatures

Task 2

Before doing the next task, you need to download Autopsy for Windows, a free forensics tool from Sleuth Kit. Download Autopsy from <https://www.autopsy.com/download/>, and then install it. For detailed information on using Autopsy, <http://sleuthkit.org/autopsy/docs/user-docs/4.17.0/>. For an explanation of the user interface, see http://sleuthkit.org/autopsy/docs/user-docs/4.17.0/ui/layout_page.html.

In the following steps, you analyze George Montgomery's USB drive. The first task is to configure Autopsy for a new case and analyze the image file of George Montgomery's USB drive. To perform the analysis, follow these steps:

2.1 Create a "C:\Work\Chap01\Chapter" folder

2.2. Download "Ch01InChap01.dd" file in C:\Work\Chap01\Chapter folder. Start Autopsy for Windows.

2.3 In Autopsy's main window, click the Create New Case button. In the New Case Information window, enter InChap01 in the Case Name text box (see Figure 1-12), and click Browse next to the Base Directory text box. Navigate to and click your work folder. Make sure the Single-user option button is selected for Case Type, and then click Next.

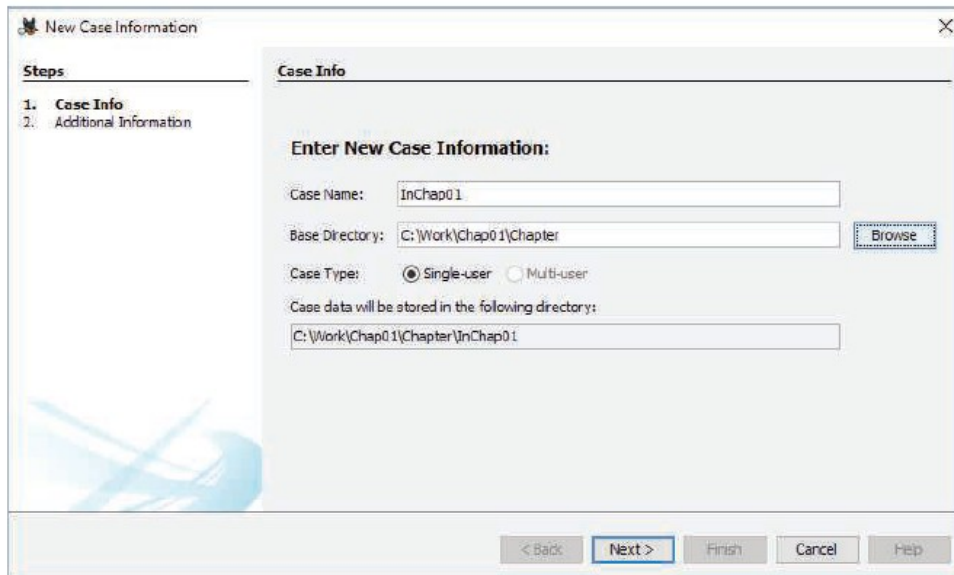


Figure 1-12 The New Case Information window

Source: www.sleuthkit.org

2.4 In the Add Data Source window, select Disk Image or VM File (see Figure 1-13), and then click Next.

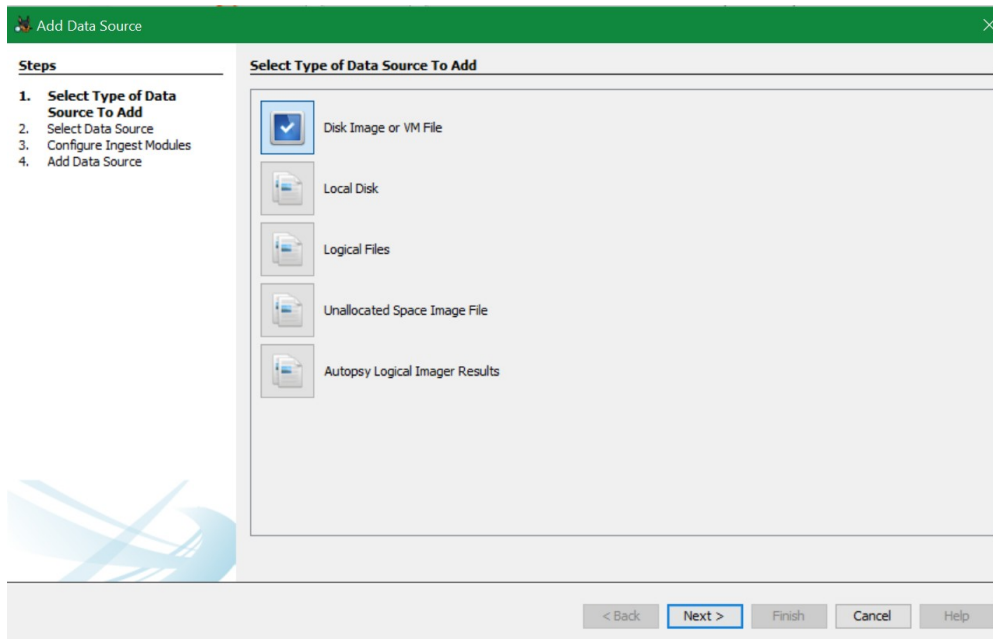


Figure 1-13

2.5 In the Select Data Source window (see Figure 1-14). Click the Browse button next to the “Browse for an image file” text box, navigate to and click your work folder and the Ch01InChap01.dd file, and then click Open. Click Next.

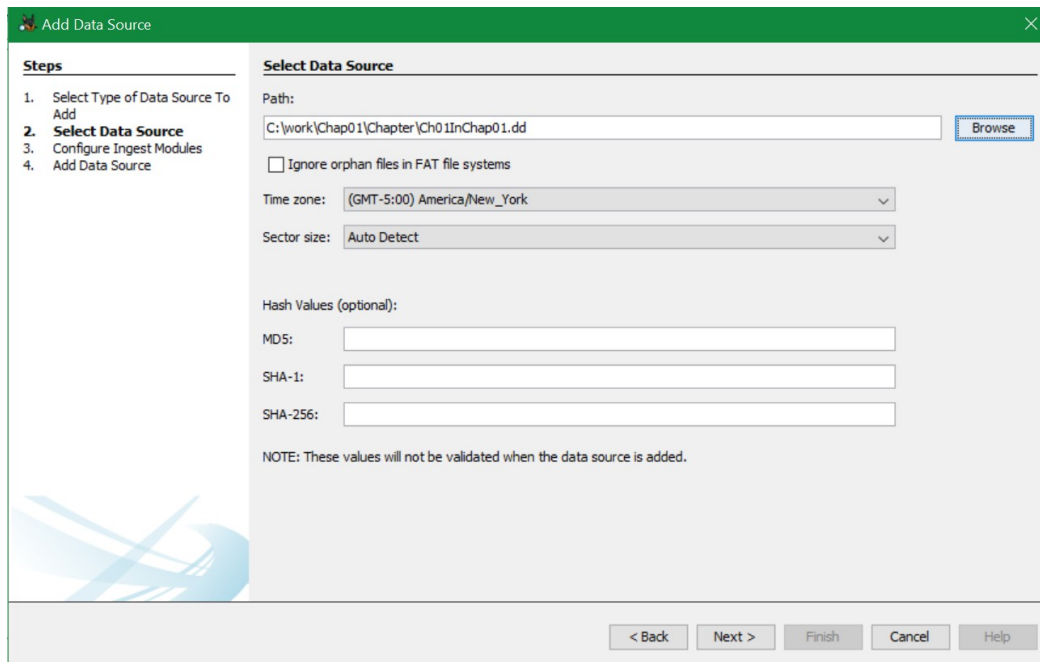


Figure 1-14

2.6 Keep the default settings in the Configure Ingest Modules window (Figure 1-14-1). Click Next and then Finish.

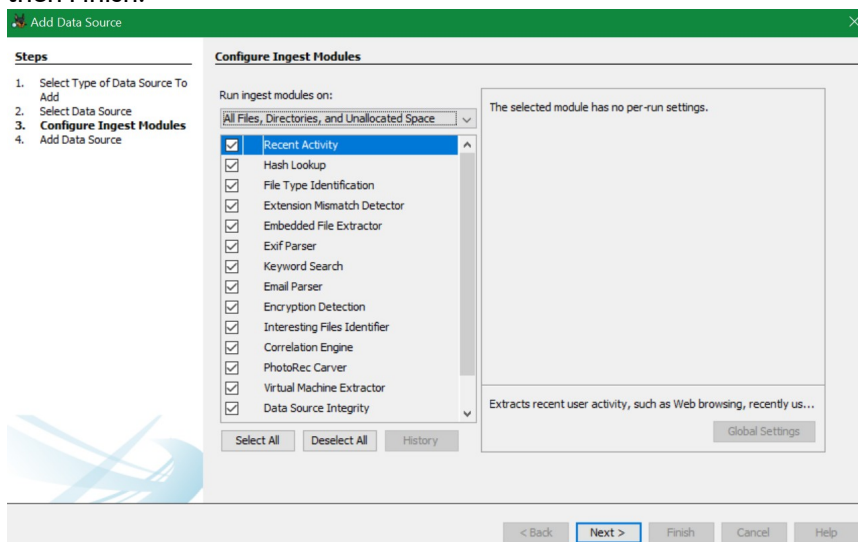
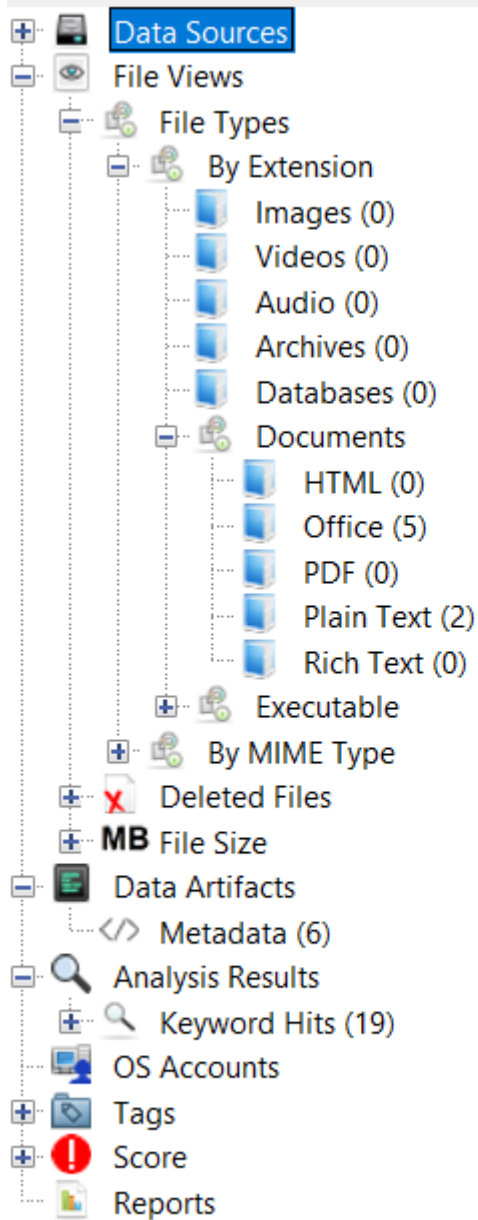


Figure 1-14-1

2.6.1. Next, follow these steps to display the contents of the acquired data: In the Tree Viewer pane on the left, click to expand Views, File Types, By Extension, and Documents (see Figure 1-15).



2.6.2. Under Documents, click Office. In the Result Viewer (upper-right pane), click the first file, Billing Letter.doc, to display its contents in the Content Viewer (lower-right pane).

Table Thumbnail Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags
Billing Letter.doc				2005-12-09 06:50:28 EST	0000-00-00 00:00:00	2005-12-09 00:00:00 EST	2005-12-09 06:59:05 EST	24064	Unallocated	Unallk
Regrets.doc				2005-12-09 06:50:52 EST	0000-00-00 00:00:00	2005-12-09 00:00:00 EST	2005-12-09 06:59:09 EST	23552	Unallocated	Unallk
f0000000_13_October_2003.doc			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	24064	Unallocated	Unallk
f0000049_02_November_2003.doc			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	23552	Unallocated	Unallk
Income.xls			0	2005-12-09 06:52:18 EST	0000-00-00 00:00:00	2005-12-09 00:00:00 EST	2005-12-09 06:59:07 EST	13824	Allocated	Alloc

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of 1 Page Matches on page: - of - Match 100% Reset Text Source: File Text

13 October 2005

Laura Roper

48 Mockingbird Lane

Seattle, WA 98119

Dear Laura,

Per our conversation, you want to register a domain with the name of www.lauras_stuff.com. You have already chosen IT Connection Servers to host your website.

I have been in contact with them to get the necessary information. My fee for the registration process is \$500. Please send a check or money order to the address listed below.

George Montgomery

3467 Main Street

2.6.3. Right-click Billing Letter.doc, point to Tag File, and click Tag and Comment.

2.6.4. In the Create Tag dialog box, click the New Tag Name button shown in Figure 1-16. In the New Tag section, type Recovered Office Documents in the Tag Name text box, click OK, and then click OK again.

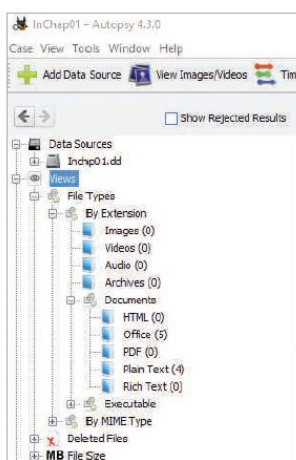


Figure 1-15 Autopsy's Tree Viewer pane

Source: www.sleuthkit.org

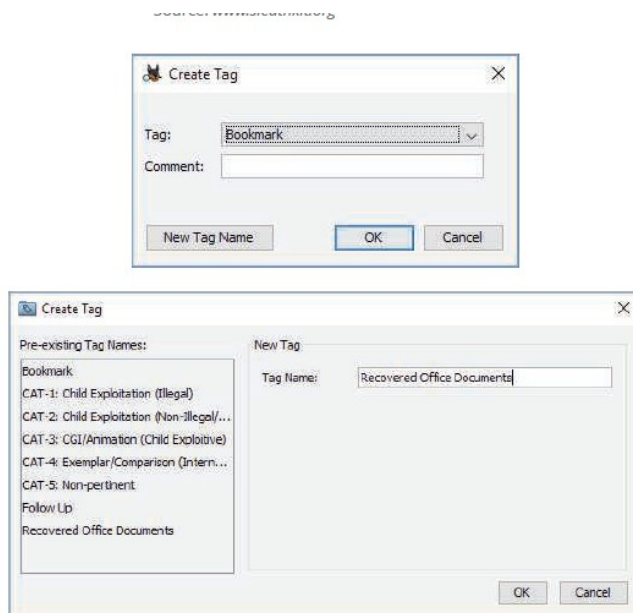


Figure 1-16 The Create Tag dialog boxes

Source: www.sleuthkit.org

2.6.5 In the Result Viewer pane, Ctrl+click Billing Letter.doc, Income.xls, Regrets.doc, f0000000.doc, and f0000049.doc to select these files, and then release the Ctrl key. Right-click the highlighted files shown in Figure 1-17, point to Tag File and then Quick Tag, and then click Recovered Office Documents.

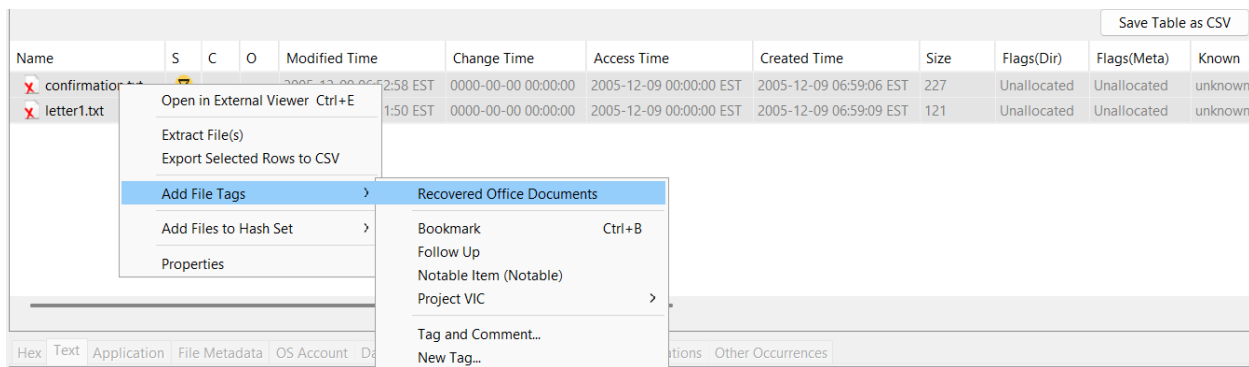
Name	S	C	O	Modified Time	Change Time	Access Time
Billing Letter.doc				2005-12-09 06:50:28 EST	0000-00-00 00:00:00	2005-12-09 00:00:00
Regrets.doc				2005-12-09 06:50:52 EST	0000-00-00 00:00:00	2005-12-09 00:00:00
f0000000_1.txt				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
f0000049_0.txt				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
Income.xls				2005-12-09 06:52:18 EST	0000-00-00 00:00:00	2005-12-09 00:00:00

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known
confirmation.txt				2005-12-09 06:52:58 EST	0000-00-00 00:00:00	2005-12-09 00:00:00 EST	2005-12-09 06:59:06 EST	227	Unallocated	Unallocated	unknown
letter1.txt				2005-12-09 06:51:50 EST	0000-00-00 00:00:00	2005-12-09 00:00:00 EST	2005-12-09 06:59:09 EST	121	Unallocated	Unallocated	unknown

2.6.6 Under Documents in the Tree Viewer pane, click Plain Text to display more recovered files.

Table Thumbnail Summary											
Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known
confirmation.txt				2005-12-09 06:52:58 EST	0000-00-00 00:00:00	2005-12-09 00:00:00 EST	2005-12-09 06:59:06 EST	227	Unallocated	Unallocated	unknown
letter1.txt				2005-12-09 06:51:50 EST	0000-00-00 00:00:00	2005-12-09 00:00:00 EST	2005-12-09 06:59:09 EST	121	Unallocated	Unallocated	unknown

2.6.7. In the Result Viewer pane, select the files listed in Step 2.6.6 again, right-click the selection, point to Add File Tag File and then click Follow Up. Leave Autopsy running for the next activity.



The next step is analyzing the data and searching for information related to the complaint. Data analysis can be the most time-consuming task, even when you know exactly what to look for in the evidence. The method for locating evidentiary artifacts is to search for specific known data values. Data values can be unique words or nonprintable characters, such as hexadecimal codes. There are also printable character codes that can't be generated from a keyboard, such as the copyright (©) or registered trademark (™) symbols. Many digital forensics programs can search for character strings (letters and numbers) and hexadecimal values, such as 0xA9 for the copyright symbol or 0xAE for the registered trademark symbol. All these searchable data values are referred to as “keywords.”

With Autopsy, you can search for keywords of interest in the case. For this case, you need to find any files associated with George Montgomery. Follow these steps to search for any reference to the name “George”:

- 2.6.8 Click the Keyword Search button at the far upper right, type George in the text box (see Figure 1-18), and then click Search.”

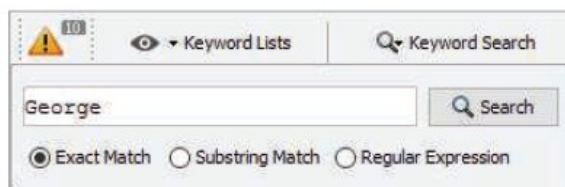


Figure 1-18 Entering a keyword search term

Source: www.sleuthkit.org

- 2.6.9 In the Result Viewer pane, a new tab named Keyword search 1 opens. Click each file to view its contents in the Content Viewer (see Figure 1-19). Look for files containing the name “George.”

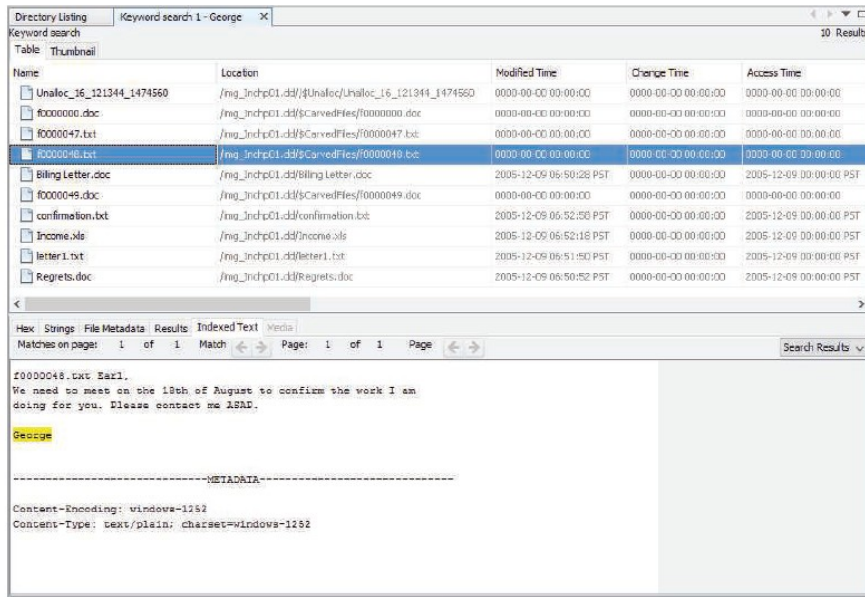


Figure 1-19 Viewing the results of searching for the keyword “George”

2.7.0 Click the Keyword Lists button at the far upper right, click the Email Addresses check box, and then click Search.

Name	Keyword Preview	Location	Modified Time	Change Time
Unalloc_4_121344_1474560	hyperlink mailto:«george.montgomery@nowhere.com»	/img_Ch01InChap01.dd/Unalloc/Unalloc_4_121344_1474560	0000-00-00 00:00:00	0000-00-00 00:00:00
Billing Letter.doc	1212 or email me at «george.montgomery@nowhere.cc	/img_Ch01InChap01.dd/Billing Letter.doc	2005-12-09 06:50:28 EST	0000-00-00 00:00:00
f0000000_13_October_2003.doc	1212 or email me at «george.montgomery@nowhere.cc	/img_Ch01InChap01.dd/CarvedFiles/1/f0000000_13_Oc	0000-00-00 00:00:00	0000-00-00 00:00:00
f0000049_02_November_2003.doc	1212 or email me at «george.montgomery@nowhere.cc	/img_Ch01InChap01.dd/CarvedFiles/1/f0000049_02_Nc	0000-00-00 00:00:00	0000-00-00 00:00:00
Regrets.doc	1212 or email me at «george.montgomery@nowhere.cc	/img_Ch01InChap01.dd/Regrets.doc	2005-12-09 06:50:52 EST	0000-00-00 00:00:00

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of 1 Page Matches on page: 1 of 10 Match 100% Reset Text Source: Search Results

HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
Regards,
George Montgomery
www.lauras_stuff.com
http://www.lauras_stuff.com/
george.montgomery@nowhere.com
mailto:george.montgomery@nowhere.com
Normal
Heading 1
Default Paragraph Font
Hyperlink
Amelia PhillipsE:\Course Technology\Computer Forensics\Chapter 2 files\Chapter 2 AU2(case files\in chapter\Billing Letter.doc
Amelia Phillips2C:\Chap02\case files\in chapter\Billing Letter.doc
Unknown
Times New Roman
Symbol

2.7.1 . In the Result Viewer pane, a new tab named Keyword search 2 opens. Click each file to view its contents in the Content Viewer pane and examine all e-mail addresses found in the search. Leave Autopsy running so that you can learn about more of its features in the next section.

Some Additional Features of Autopsy

If you find a file of interest that displays binary (nonprintable) data in the Content Viewer, there are six available viewing options in Autopsy: Hex, Strings, File Metadata, Results, Indexed Text, and Media. To view an unallocated sector for its hexadecimal values, click the Hex tab in the Content Viewer, as shown in Figure 1-20.

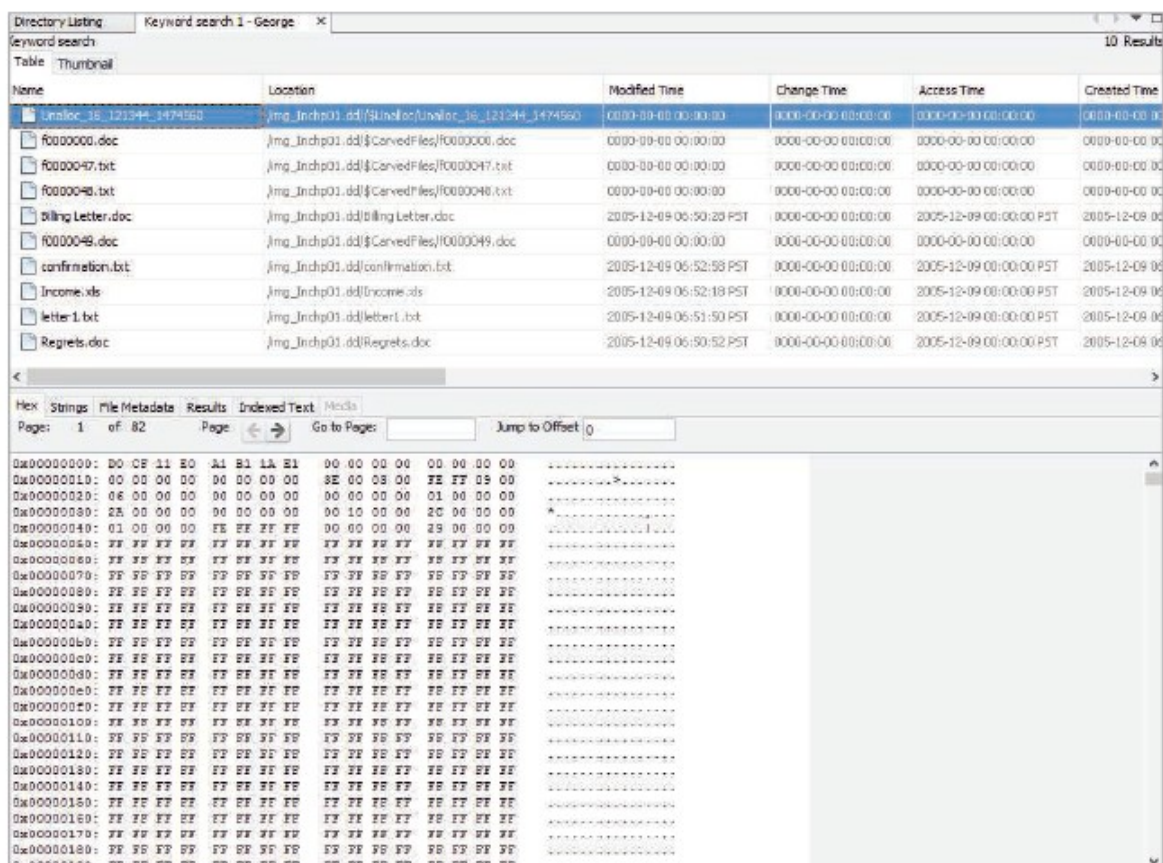


Figure 1-20 Viewing search results found in unallocated drive space

Source: www.sleuthkit.org

If you want to examine an Office file, such as an Excel spreadsheet, right-click the file and click Open in External Viewer. To use this option, you must have the external viewer's software installed on your computer. To export a file from Autopsy, select it, and then right-click the file and click Extract File. In the Save dialog box that opens, you can copy the file to a work folder.

Autopsy's Report Generator

Autopsy has several styles of reports, including a plain text file, an HTML Web page with links to artifacts, and an Excel spreadsheet. To generate a report, you can follow this general procedure:

2.7.2. If you exited Autopsy, start it again, and click Open Recent Case. Click InChap01 and then click Open in the Recent Case window. In Autopsy's main window, click the Generate Report button at the top.

2.7.3 In the Generate Report window, select the report format you want in the Report Modules section (see Figure 1-21). The Results - HTML option, for example, produces a linkable Web page with tagged artifacts, and the Files - Text option creates a plain text output file. When you're finished, click Next.

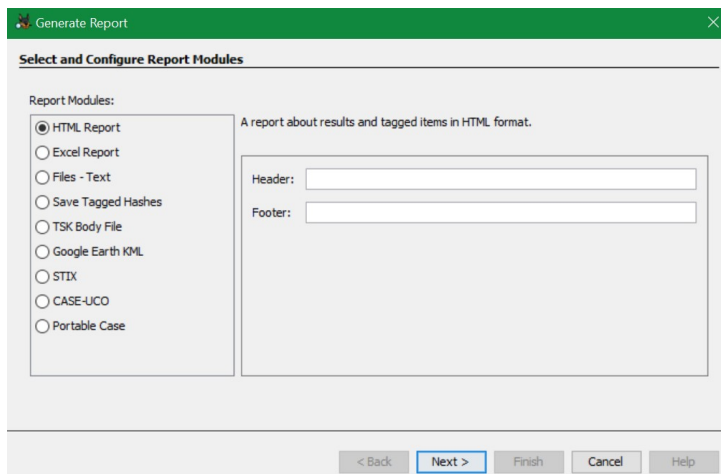
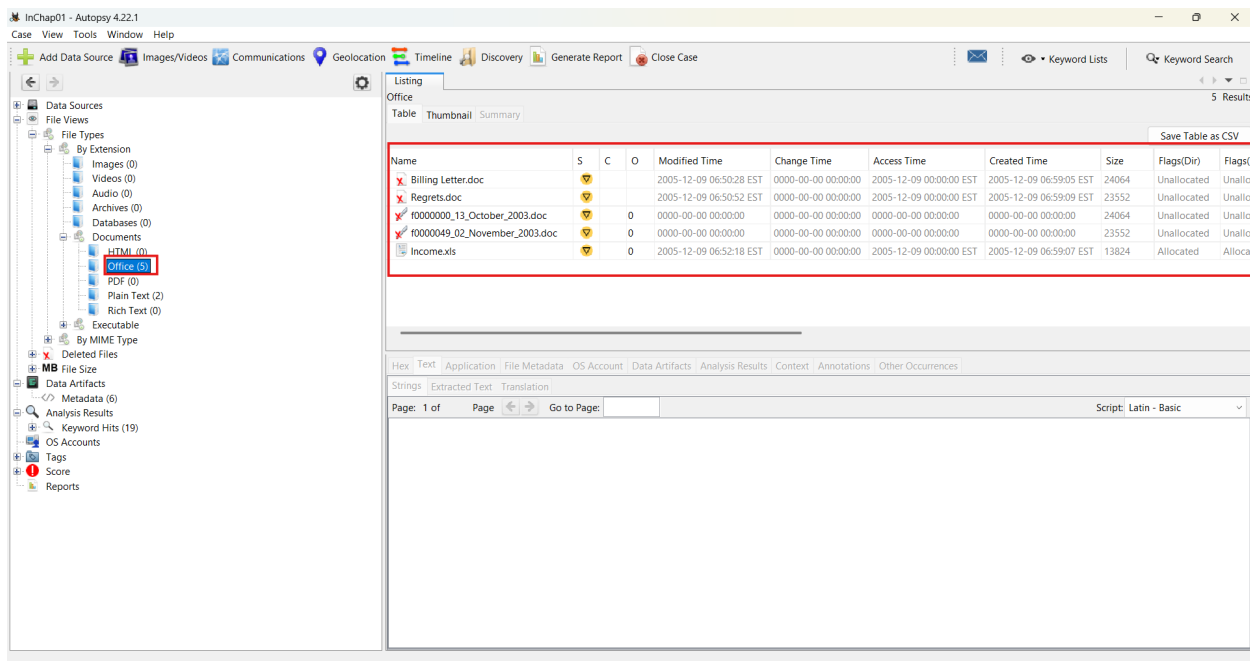


Figure 1-21

Screenshots:



InChap01 - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing

Plain Text 2 Results

Table Thumbnail Summary

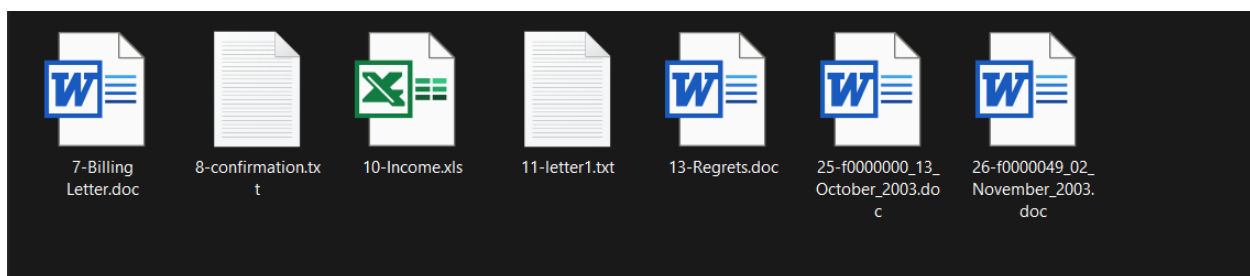
Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known
confirmation.txt				2005-12-09 06:52:58 EST	0000-00-00 00:00:00	2005-12-09 00:00:00 EST	2005-12-09 06:59:06 EST	227	Unallocated	Unallocated	unknown
letter1.txt				2005-12-09 06:51:50 EST	0000-00-00 00:00:00	2005-12-09 00:00:00 EST	2005-12-09 06:59:09 EST	121	Unallocated	Unallocated	unknown

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of Page Go to Page: Script: Latin - Basic



(Exported files)

Generate Report

×

Select and Configure Report Modules

Report Modules:

☐ HTML Report

☐ Excel Report

☒ Files - Text

☐ Data Source Summary Report

☐ Save Tagged Hashes

☐ Extract Unique Words

☐ TSK Body File

☐ Google Earth KML

☐ CASE-UCO

☐ Portable Case

A delimited text file containing information about individual files in the case.

☒ Tab delimited

☐ Comma delimited

< Back

Next >

Finish

Cancel

Help

(Report)

Task 3

The case in this project involves a suspicious death. Joshua Zarkan found his girlfriend's dead body in her apartment and reported it. The first responding law enforcement officer seized a USB drive. A crime scene evidence technician skilled in data acquisition made an image of the USB drive with FTK Imager and named it C1Prj01.E01. Following the acquisition, the technician transported and secured the USB drive and placed it in a secure evidence locker at the police station. You have received the image file from the detective assigned to this case. He directs you to examine it and identify any evidentiary artifacts that might relate to this case. To process this case, follow these steps to evaluate what's on the image of the USB drive:

Download C1Prj01.E01 (week2) from DC Connect

3.1. Start Autopsy for Windows and click the Create New Case icon. In the New Case Information window, enter C1Prj01 in the Case Name text box, and click Browse next to the Base Directory text box. Navigate to and click your work folder, and then click Next.

3.2. In the New Case Information window, type C1Prj01 in the Case Number text box and your name in the Examiner text box, and then click Finish.

3.3. In the Add Data Source window, click Image File or VM File and click next, Click the Browse button next to the "Browse for an image file" text box, navigate to and click your work folder and the C1Prj01.E01 file (Figure 1-22), and then click Open. Click Next.

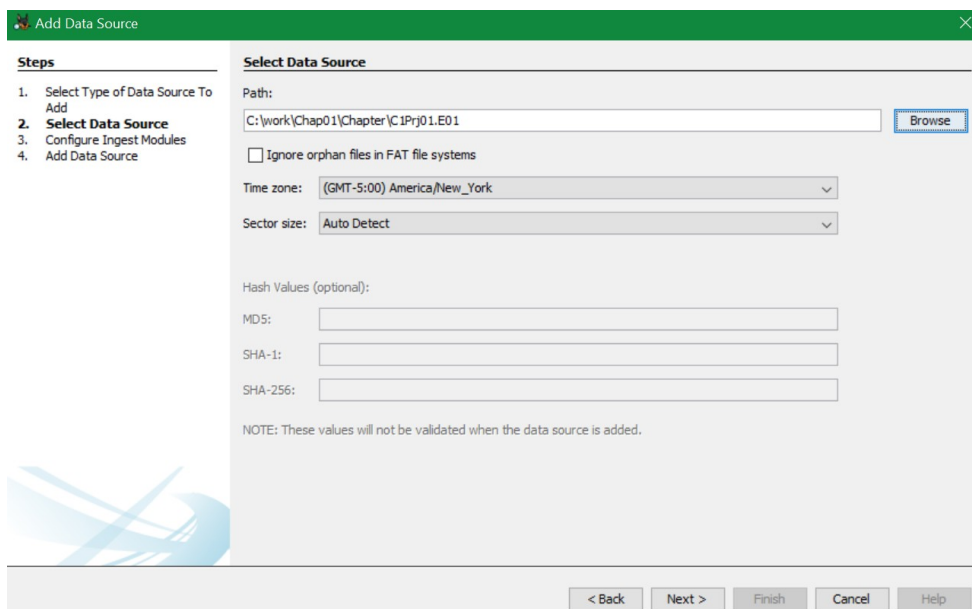
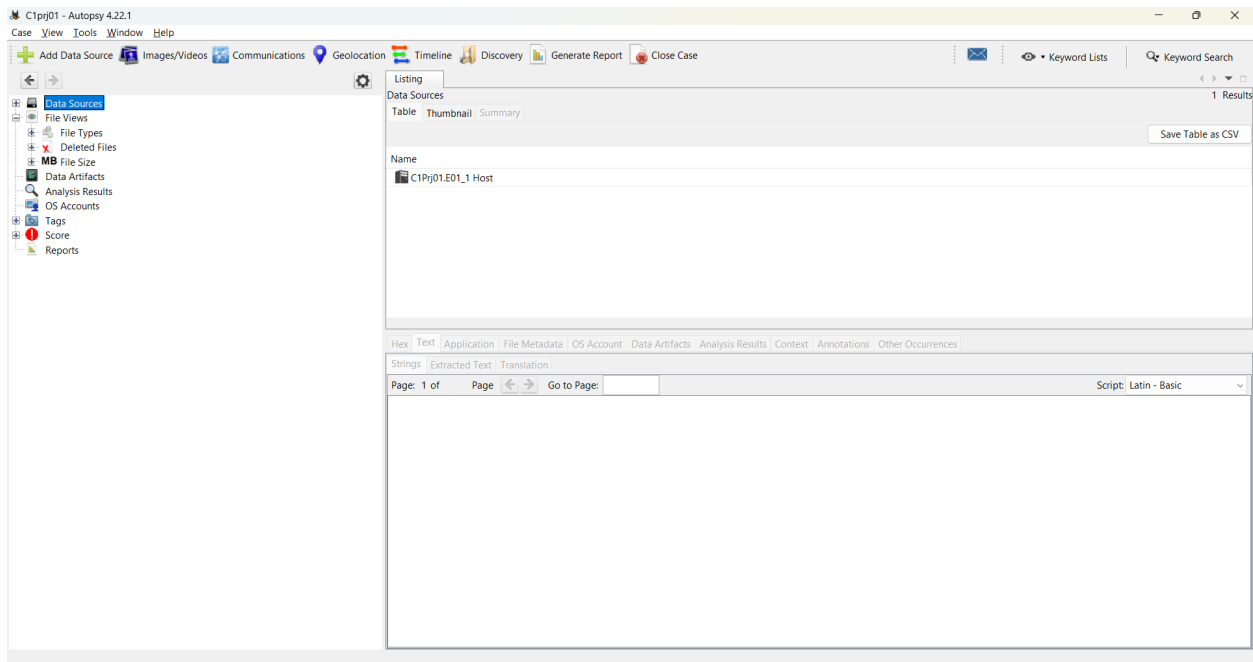


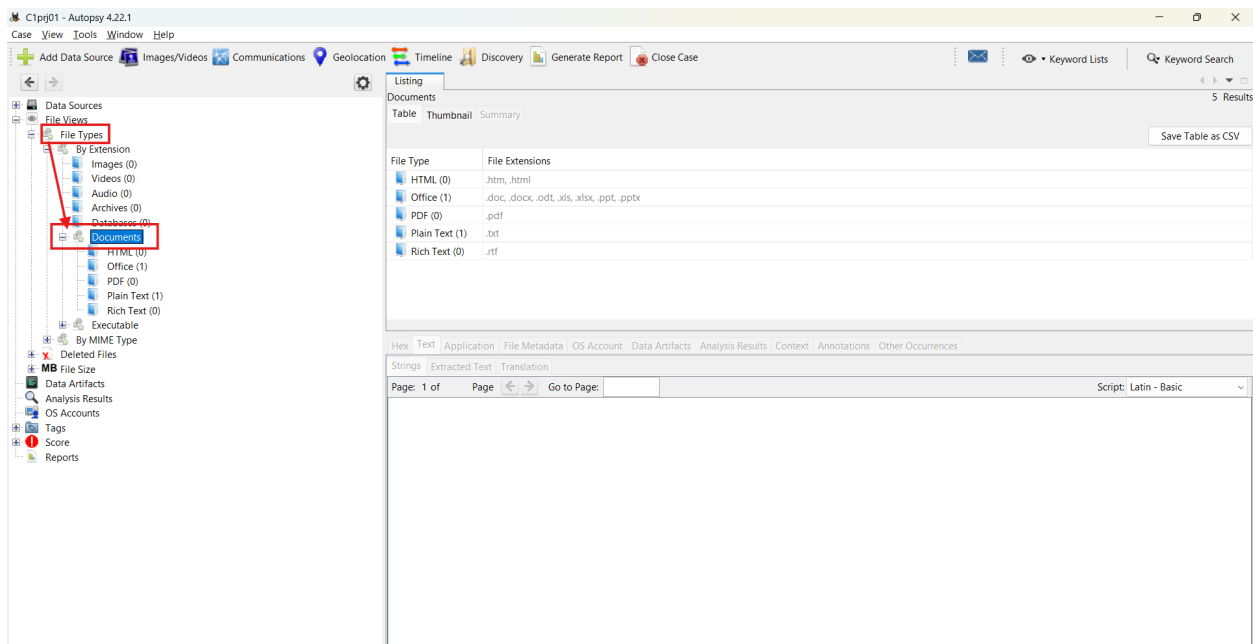
Figure 1-22

3.4. In the Configure Ingest Modules window, click Select All. Click Next and then Finish.



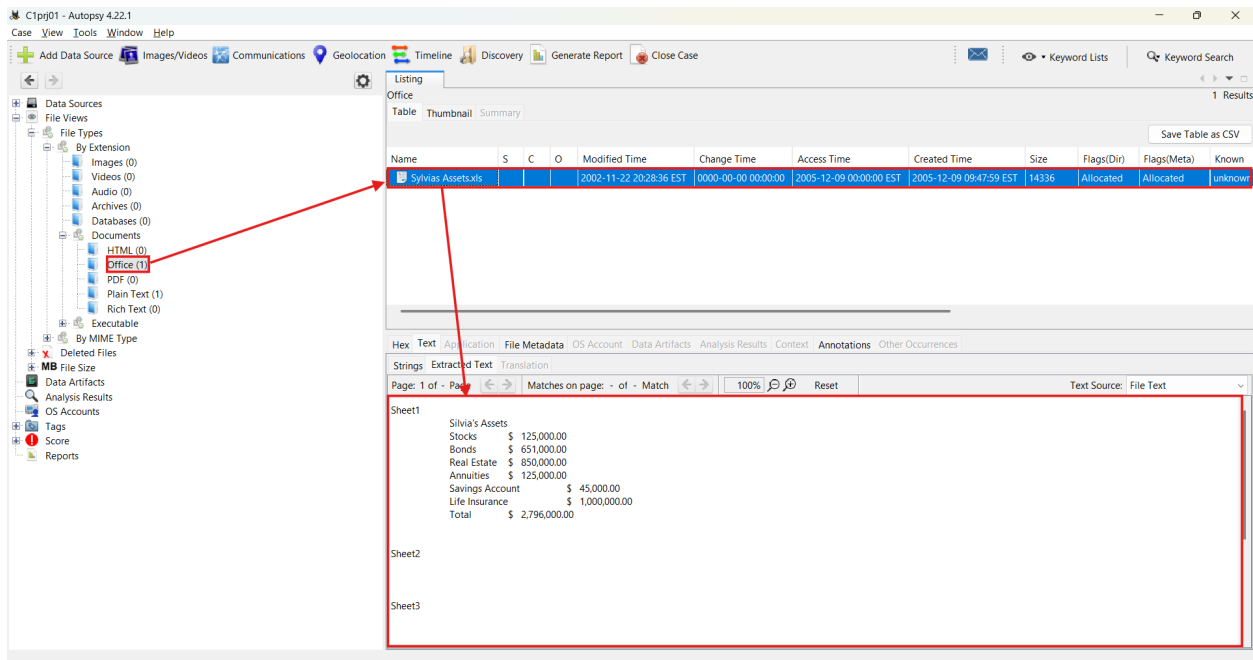
(Project Created)

3.5. In the Tree Viewer pane, expand Views, File Types, By Extension, and Documents.

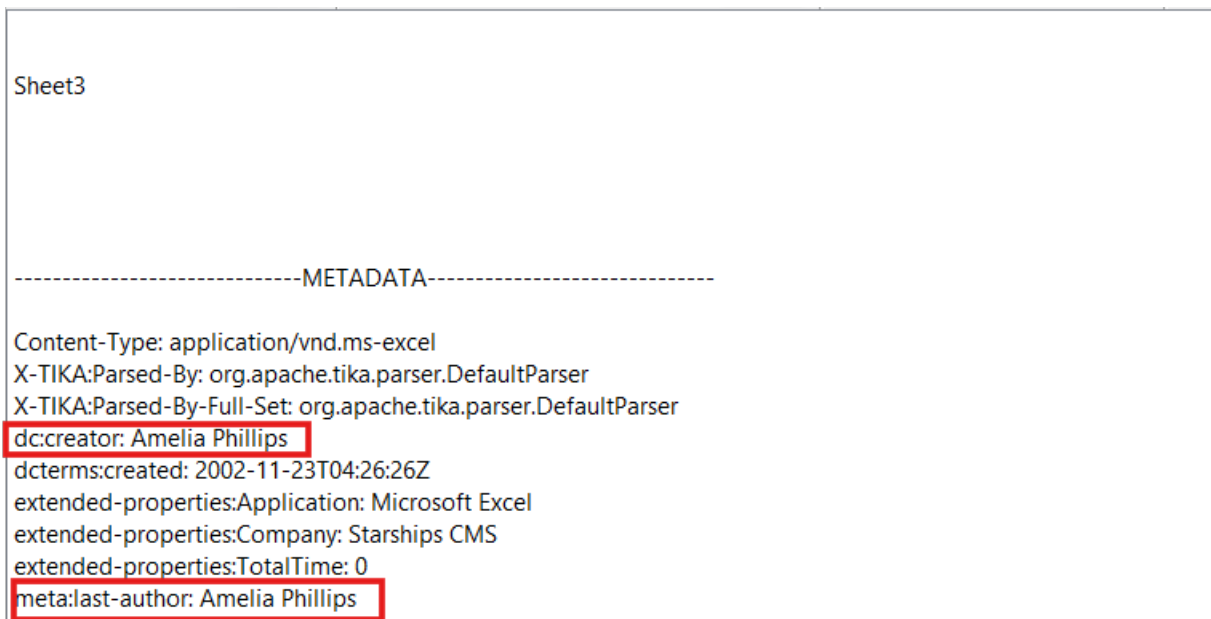


(In Document Section)

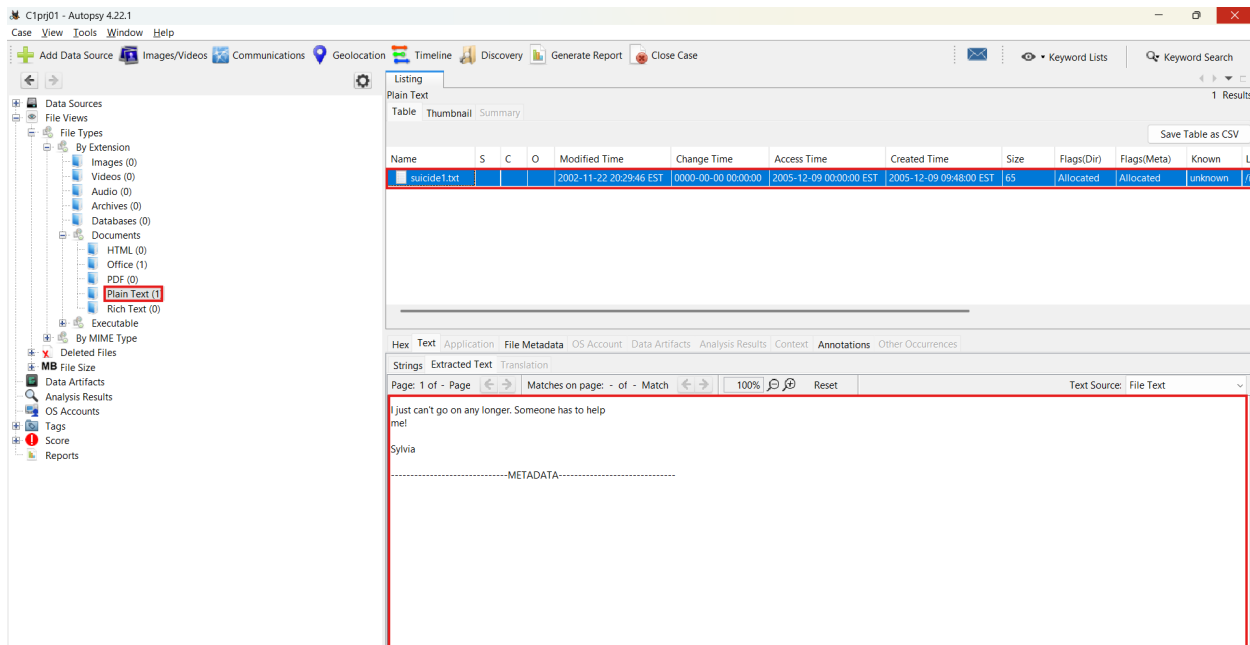
3.6. Examine each subfolder under Documents. Determine which folder might contain files of interest to this case.



(In Office sections there's one file named "Sylvias Assets" which contains detailed Sylvia's Assets with valuations)



(Metadata Shows Amelia Philips made this and it was last edited by her)



(Another file is found in plain text section named “suicide1.txt”. No metadata found)

Metadata

Name: /img_C1Prj01.E01/suicide1.txt
 Type: File System
 MIME Type: text/plain
 Size: 65
 File Name Allocation: Allocated
 Metadata Allocation: Allocated
 Modified: 2002-11-22 20:29:46 EST
 Accessed: 2005-12-09 00:00:00 EST
 Created: 2005-12-09 09:48:00 EST
 Changed: 0000-00-00 00:00:00
 MD5: Not calculated
 SHA-256: Not calculated
 Hash Lookup Results: UNKNOWN
 Internal ID: 7

From The Sleuth Kit istat Tool:

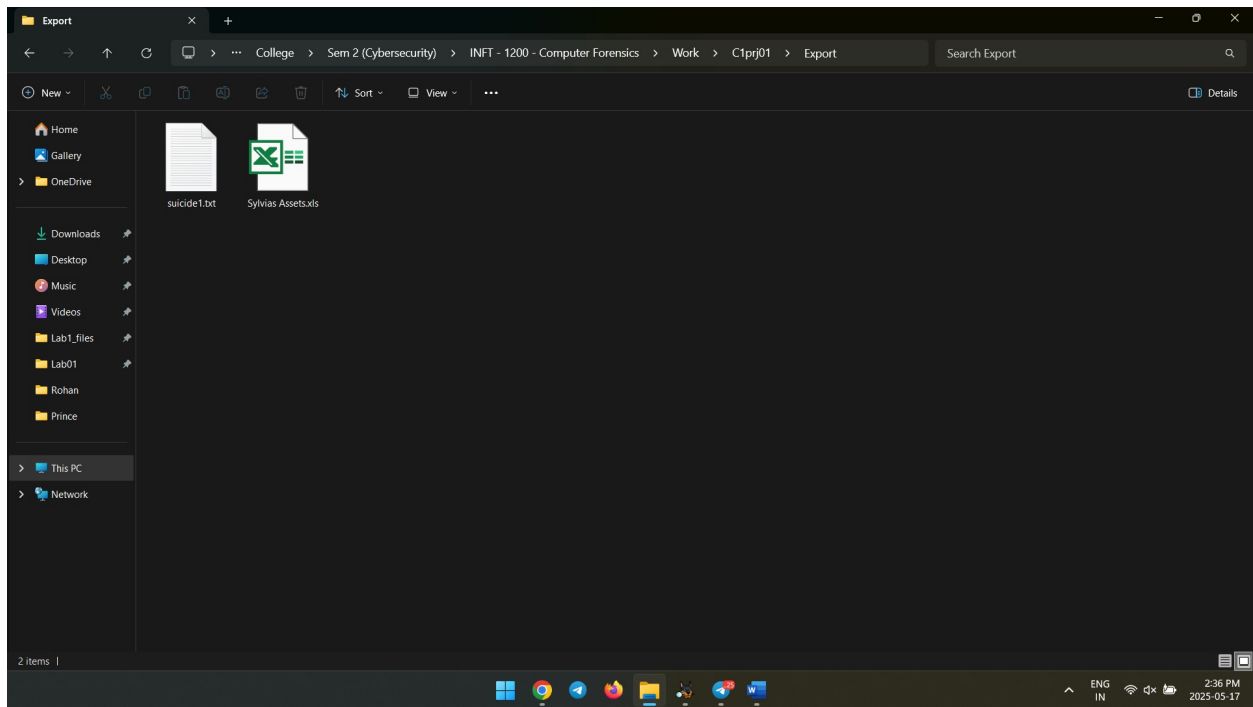
Directory Entry: 7
 Allocated
 File Attributes: File, Archive
 Size: 65
 Name: SUICIDE1.TXT

Directory Entry Times:
 Written: 2002-11-22 20:29:46 (Eastern Standard Time)
 Accessed: 2005-12-09 00:00:00 (Eastern Standard Time)
 Created: 2005-12-09 09:48:00 (Eastern Standard Time)

Sectors:
 Starting address: 61, length: 1

(Metadata shows last updated date and time with last access time)

3.7. If you found any files related to the case, select the files as a group, right-click the selection, and click Extract File(s). In the Save dialog box, click Save to save the files automatically in Autopsy’s case subfolder: **Work \Chap01\Projects\C1Prj01\Export**.



(Extracted files)

Reflective statements (end-of-exercise):

You should reflect on these questions:

1. Briefly describe the triad that makes up computer security.
 - The CIA Triad stands for the three basic principles of computer security: confidentiality, integrity, and availability.
 - Confidentiality makes sure that information is only accessible to those who have the required authorization. This goal is generally realized through approaches like applying access controls and following the principle of least privilege.
 - Integrity requires information to be accurate, consistent, and alterable only by authorised people for legitimate ends.
 - Availability ensures the perpetual availability of systems, resources, and information to entitled people and reduces the occurrence of unintentional discontinuity or unavailability.
 - These three essential elements work together to safeguard digital resources and promote trust in information systems.
2. What are some of the most common types of private-sector computer crime?
 - Common computer crimes in the private sector include phishing assaults, which deceive employees into disclosing critical information; ransomware, which encrypts firm data and demands payment; and insider threats, which include the misuse of access by employees or contractors. Other common crimes include data breaches, which expose confidential customer or business data, intellectual property theft, such as stealing trade secrets or proprietary code, financial fraud through digital transaction manipulation, and denial-of-service (DoS) attacks, which disrupt access to critical services. These accidents can have a significant impact on an organization's finances, operations, and reputation.
3. Why should the evidence media be write-protected?
 - Media evidence should be write-protected in order to keep the information intact and prevent any alteration when under analysis. This is critical because even slight, accidental alterations may taint the integrity of the evidence and render it inadmissible during court proceedings. Write-protection guarantees the integrity of the information, allows for a proper chain of custody, and guarantees that forensic experts can conduct a legitimate and trustworthy investigation. It also adds assurance to the results by showing that the evidence has not been changed since it has been collected.

References

<https://asecuritysite.com/forensics/magic>